

Structural transfer using EDAs: An application to multi-marker tagging SNP selection

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Abstract—In this paper we investigate the question of transfer learning in evolutionary optimization using estimation of distribution algorithms. We propose a framework for transfer learning between related optimization problems by means of structural transfer. Different methods for incrementing or replacing the (possibly unavailable) structural information of the target optimization problem are presented. As a test case we solve the multi-marker tagging single-nucleotide polymorphism (SNP) selection problem, a real world problem from genetics. The introduced variants of structural transfer are validated in the computation of tagging SNPs on a database of 1167 individuals from 58 human populations worldwide. Our experimental results show significant improvements over EDAs that do not incorporate information from related problems.

I. INTRODUCTION

The most common procedure when applying evolutionary algorithms (EAs) to real world problems is running a single EA to the problem at hand and eventually tuning the EA parameters to improve its performance. However, in many situations the end user is posed with the task of solving a set of similar optimization problems or even a set of dissimilar but somewhat related problems. In these situations, the question of how to reuse problem information in order to improve the efficiency of the EAs arises.

Model-based EAs are algorithms able to learn and use characteristic features of the search space during the solution of the optimization problem. Perhaps the best known class of these algorithms that apply modeling during the evolutionary search are estimation of distribution algorithms (EDAs) [1], [2], [3]. The distinguished feature of EDAs is that statistical techniques are used to learn the regularities in the search and these regularities are represented using probabilistic models.

For several problems with important variable interactions, EDAs guarantee a more efficient search than EAs based on blind genetic operators. They are also able to capture structural information about the problem being solved that can reveal unknown information about the problem characteristics, in particular its structure.

EDA modeling ability makes these algorithms excellent candidates for implementing information transfer between similar problems. Some recent EDA research has focused on how to extract problem information from the probabilistic

models learned by EDAs and how to reuse this information in new related problems. This paper is devoted to the second question. We propose different alternatives for transferring information between problems. In particular, we focus on transferring structural information between problems that may have different number of variables and where the relationships between the variables are only partially preserved when moving between problems. To illustrate the gains achieved by our proposal, we focus on an important problem from genetics. The multi-marker tagging SNP selection problem is addressed for a set of 58 human populations with different genetic proximity. We have previously shown [4] that EDAs are state of the art algorithms for this problem. Results presented in this paper show that structural transfer can further and significantly improve previous results achieved with EDAs.

The paper is organized as follows: In the next section we analyze the transferability problem in EAs. Section III presents the multi-marker tagging SNP selection problem. Section IV describes Tree-EDA and Tree-EDA^r, the EDAs used in this paper. Our proposals for knowledge transference in the multi-marker tagging SNP selection problem are described in Section V. In Section VI we briefly discuss related work on transfer learning and the use of a priori information for EDAs. Section VII presents the experimental framework and the numerical results of our simulations. The conclusions and some ideas for future work are given in Section VIII.

II. TRANSFERABILITY IN EVOLUTIONARY ALGORITHMS

A. Transfer learning, domain adaptation and multi-task learning

Ideal machine learning methods should be able to retain previously learned knowledge and reuse it across a variety of application contexts. From a general point of view, *transfer learning* [5] is related to how to utilize those data from different time periods or domains to help the current learning task [6]. It has been mainly applied to classification problems but also in regression [7], clustering [8], reinforcement learning [9] and optimization [10]. Transfer learning is usually an umbrella for more specific machine learning techniques such as domain adaptation, multitask learning and sample selection bias. This is so because in all these techniques knowledge or information

is transferred across domains or tasks. However, there are some differences between these techniques that are relevant for their application in the context of EAs and therefore we analyze them in more detail.

Domain adaptation [11] usually addresses classification problems for which the training instances are governed by a distribution that is allowed to differ arbitrarily from the test distribution. Labeled data from one domain is used to train a model that is expected to maximize accuracy in a target domain. This type of problems are also referred to as classification under covariate shift [12].

Multi-task learning [13], [14] refers to the problem of learning multiple tasks simultaneously. These tasks may share different degrees of similarity. An important issue in multi-task learning is how to define the relatedness between tasks [15]. A difference between multi-task learning and transfer learning is that while in the former the goal is to solve all tasks, the second focuses on the results for the target domain.

B. Alternatives for transfer learning in EAs

Our general goal is to solve an optimization problem in a target domain and for that purpose we want to use information about, to some extent related, optimization problems that can be unsolved, being solved or already solved. We define this problem as *transfer learning for optimization algorithms*. Notice that while knowledge is extracted from the domain problem(s) and applied to the target problem(s) the knowledge is transferred from or to an optimization algorithm. We emphasize the role of the optimization algorithm at some point of the transfer process, either to extract the information from the optimization domain, to apply the information of the optimization domain to the target domain, or in both steps of the process. There are three main questions that should be considered:

- Which type of information (problem descriptors) to extract from the problems?
- How to define and measure the relatedness between the problems?
- How to transfer the available information to the solution of the target optimization problem?

All the above questions are related. The type of information extracted from the problem will influence the way we measure the similarity or relatedness between the problems. Similarly, the type of problem descriptors available will constrain the range of alternatives for reusing the information.

We identify the following alternatives for transfer learning for optimization:

- Transfer of solutions: The modification or reuse of a solution from the domain optimization problem in the target problem.
- Structural transfer: The use of a structural description of the domain problem as an approximate structural description of the target problem.
- Behavioral or algorithmic transfer: The reuse of the algorithm behavior in the domain problem as a way to

bias the behavior of the algorithm in the target domain. This can be done by modifying the parameters of the optimization algorithm such as to emulate the behavior exhibited while successfully solving the domain problem in order to eventually obtain similarly good solutions for the target problem.

- Combined transfer: Any combination of the previous three alternatives.

We will focus on structural transfer and consider the situation in which some information from a set of related problems is available. In Section V our proposals for transfer learning in the context of the multi-marker tagging SNP selection problem are discussed.

III. MULTI-MARKER TAGGING SNP SELECTION PROBLEM

In the human genome there are about 10 million sites where individuals differ by a single nucleotide. These sites are called single nucleotide polymorphisms (SNPs). An *allele* is an alternative form of a gene or SNP, or another type of variant. Most SNPs are *bi-allelic*, i.e. they appear as having only two possible nucleotides. A *haplotype* is a combination of alleles at multiple linked sites on a single chromosome, all of which are transmitted together. A *haplotype block* is a region containing strongly associated SNPs.

A chromosome carrying a particular allele of a given SNP has a high probability of carrying a particular allele of another SNP close to the first one. Thus, an allele frequency difference in the second SNP can manifest itself as an allele frequency difference in the first SNP. The non-random association of alleles at two or more sites on the same chromosome is called *linkage disequilibrium* (LD) and this relationship is often measured by the correlation coefficient r^2 between SNPs. A *tagging* or *tag* SNP is a representative SNP with high LD to other (*tagged*) SNPs. The identification of tagging SNPs sets is still a more affordable alternative to genome-wide analysis in disease-gene association studies and therefore the computation of tagging SNPs is an important problem in Bioinformatics.

Let D be a data set consisting of m haplotypes, $\mathcal{H} = \{h_1, \dots, h_m\}$, each with n different SNPs, $S = \{s_1, \dots, s_n\}$. The set D can be viewed as an $m \times n$ matrix. For simplicity of presentation, we assume in our analysis that each of the SNPs is bi-allelic. Let (A, a) and (B, b) respectively represent the two possible alleles for two different SNPs s_i and s_j in D . The correlation coefficient r_{ij}^2 measures the similarity correlation between the two SNPs.

We say that SNP s_i tags SNP s_j if their correlation coefficient r_{ij}^2 exceeds some threshold r_{min}^2 . We call $T_{sin} \subseteq S$ a single-marker valid tag of S if $\forall s_j \in S, \exists s_i \in T_{sin}$ such that $r_{ij}^2 \geq r_{min}^2$, i.e., for each SNP s_j in S there exists a SNP s_i in T_{sin} such that the correlation between both SNPs is higher than the threshold r_{min}^2 , and therefore it is expected that one SNP can be predicted from the other. The correlation coefficient r_{ij}^2 can be generalized to groups of SNPs. In particular, we will denote the correlation coefficient between a couple of SNPs s_i and s_j with respective alleles

(A, a) and (B, b) and a SNP s_k with possible alleles (C, c) as $r_{\{i,j\},k}^2$.

We say that a subset $T_{mul} \subseteq S$ is a multi-marker valid tag of S if for all SNP s_i in S there exists a subset of SNPs $T^* \subseteq T_{mul}$ such that the correlation between the SNPs in T^* and s_i is higher than the threshold, i.e., $r_{T^*,i}^2 > r_{min}^2$.

The problem of finding the smallest single-marker tagging set is the problem of finding the smallest set $T_{sin} \subseteq S$ that is a valid tag of S . Similarly, the problem of finding the smallest multi-marker tagging set is the problem of finding the smallest set $T_{mul} \subseteq S$ that is a valid multi-marker cover of S .

In this paper we focus on the second class of problems. We further constrain the set of multi-marker tagging sets to those where the tagging set of each SNP is formed by at most two tagging SNPs.

A. Search space and problem representation

Given a set S of n SNPs, the search space is composed of the valid multi-marker subset of S . However there may exist SNPs that are not covered by any single or pair of tagging SNPs. These SNPs can be only self-tagged, so we call them fixed SNPs. Given that fixed SNPs should appear in each valid multi-marker set of S we do not consider them in the search and therefore we need to carry out a search inside a set of n' SNPs.

We codify a possible solution to the problem (valid multi-marker set) as a binary n' length vector $\mathbf{x} = (x_1, \dots, x_{n'})$. Variable X_i will represent whether the i th SNP is part of the tagging set ($x_i = 1$), or it is tagged ($x_i = 0$).

The final solution comprises all fixed SNPs and those found during the search.

B. Fitness function

For implementational reasons, the minimization of the number of tagging SNPs is transformed in the maximization of Equation (1), where each solution $\mathbf{x}$ satisfies that all the non-tagging SNPs are covered by another single or pair of tagging SNPs. Hence given a solution $\mathbf{x}$, $f(\mathbf{x})$ gives back n' minus the number of SNPs in the valid multi-marker set.

$$f(\mathbf{x}) = n' - \sum_{i=1}^{n'} x_i \quad (1)$$

IV. EDA APPROACHES

A. Probabilistic model

The EDA of choice uses a probabilistic model that captures bivariate dependencies between the variables. This probabilistic model is based on a tree structure where each variable may depend on, at most, another variable, which is called the parent. A probability distribution $p_{Tree}(\mathbf{x})$ that is conformal with a tree is defined as:

$$p_{Tree}(\mathbf{x}) = \prod_{i=1}^n p(x_i | pa(x_i)) \quad (2)$$

where $Pa(X_i)$ is the parent of X_i in the tree, and $p(x_i | pa(x_i)) = p(x_i)$ when $Pa(X_i) = \emptyset$, i.e. X_i is the root of the tree. The distribution $p_{Tree}(\mathbf{x})$ itself will be called a tree model when no confusion is possible.

B. Tree-EDA and Tree-EDA^r

The construction of the tree structure from data implies the detection of the most important bivariate interactions between the variables. We use the analysis of the mutual information between variables to construct the models as described in [16]. The main steps of Tree-EDA are shown in Algorithm 1.

Algorithm 1: **Tree-EDA**

```

1   $D_0 \leftarrow$  Generate  $M$  individuals randomly
2   $l = 1$ 
3  do {
4     $D_{l-1}^s \leftarrow$  Select  $N \leq M$  individuals from  $D_{l-1}$ 
      according to a selection method
5    Compute the univariate and bivariate marginal
      frequencies  $p_i^s(x_i | D_{l-1}^s)$  and  $p_{i,j}^s(x_i, x_j | D_{l-1}^s)$  of
       $D_{l-1}^s$ 
6    Calculate the matrix of mutual information using
      bivariate and univariate marginals.
7    Calculate the maximum weight spanning tree from
      the matrix of mutual information.
8    Compute the parameters of the model.
9     $D_l \leftarrow$  Sample  $M$  individuals (the new population)
      from the tree and add elitist solutions.
10 } until A stop criterion is met

```

In [4], it is shown that the use of a priori information about the problem structure can improve the quality of the solutions for the multi-marker tagging SNP selection problem. There is information about the correlations between the SNPs that can be incorporated to the model.

We use a variant of the tree learning algorithm that constrains the calculation of the mutual information to those pairs of variables corresponding to SNPs that have some potential type of tagging relationship, i.e., they belong to a pair (tagging-tagged) or to a triple (tagging,tagging,tagged) of SNPs. The assumption is that any other pairwise relationship between SNPs is not relevant for the search of the optimal solutions. Step 6 of the tree learning algorithm is modified so that the mutual information between two variables X_i and X_j is computed only if their corresponding SNPs, s_i and s_j are involved in a tagging relationship.

The variant of Tree-EDA that restricts the interactions represented by the tree structure to interacting pairs of variables is called Tree-EDA^r. This algorithm reduces the computational time of the learning step in the EDA and also improves the reached solutions.

It must be taken into account that not all the sampled solutions are feasible, in the sense that there are binary vectors that represent situations in which one or more SNPs could be not covered. To keep the search in the space of feasible solutions, we implement a repairing procedure that enforces the solutions feasibility.

This procedure is applied during the evaluation step. The algorithm starts by checking whether $\mathbf{x}$ is a feasible solution. For efficiency reasons, the verification is done by firstly taking into account the single tagging SNPs and then the pairs of tagging SNPs. If the set of non-tagged SNPs is not empty (i.e. the solution is unfeasible), each of the non-tagged SNPs becomes tagged by transforming some of them into tagging SNPs (x_i from 0 to 1). The repairing procedure is thought to set as few tagging SNPs as possible. It finishes when all the SNPs are tagged. For more details on this procedure see [4].

V. IMPLEMENTATION OF THE TRANSFER IDEAS

A. Transferability of tagging SNPs

Usually, most tagging SNP sets are defined in a single population and applied only in populations that are closely related [17]. The problem of tagging SNP transferability refers to the question of reusing the tagging SNP set found for a given population in other populations. The relevance of the problem is given by different reasons: Typing a single tagging SNP in several different populations can be less expensive than typing a distinct SNP in each population [17]. Also, the availability of exhaustive SNP databases is still scarce. Therefore, it makes sense to use the available SNP databases to find candidate tagSNP sets that could be applied to other populations different to those from which the database has been constructed. The transferability of tagging SNPs has been extensively investigated [18], [19], [20].

If we consider each human population as a different problem domain, the computation of tagging SNPs is a paradigmatic example of an optimization problem where solutions (sets of tagging SNPs) for different domains may be related. The degree of relatedness between solutions will be associated with the genetic similarity between populations. This is the reason why, in general, tagging SNPs found for a population are expected to be “portable” to other populations with similar patterns of genetic variation. Studies on SNP transferability are mostly focused on determining the extent to which a given set of tagging SNPs is transferable to other populations.

B. Transferability for tagging SNP computation

The main idea behind our approach will be to use the structural information from one or more tagging SNP problems to construct an approximate matrix of interactions that serves to bias model construction in Tree-EDA^r. It should be noted that the information available from different domains can be partial, in the sense that different populations usually share only a subset of SNPs. Therefore a number of subproblems should be solved for effective transfer.

We assume that a set of SNP datasets corresponding to different populations is available. There is one dataset for each population. From each dataset i we have the SNPs involved in the optimization problem i and a matrix of interactions between them. Different SNP datasets only share a number of common SNPs and the interactions between these SNPs are not identical. For a given SNP dataset i , we will use this initial information for implementing the transfer process.

In the solution of problem i , the relevant information that is extracted from the related problems is always summarized in an aggregation matrix that will contain the interactions passed to Tree-EDA^r as described in Section IV-B. Four variants for creating the aggregation matrix have been implemented:

- Aggregated interactions ($Tn1$): We create a matrix of interactions for problem i by combining the interactions between any pair of SNPs in problem i that were detected in any of the other SNP datasets. Basically, we are enlarging the set of potential interactions between the variables of problem i by using information from the other problems. The information about the interactions in problem i is also included in this matrix.
- Pure transfer ($Tn2$): This implementation is identical to $Tn1$ but information about the interactions in problem i is not included in the aggregation matrix. This situation emulates the case where we do not know the interactions between the variables for the target problem but this information is available from related problems.
- Limited information, overlapping variables ($Tn3$): Here, information about only a very reduced subset of the SNP databases is used. The *most related* problems to problem i are selected as those that share the largest set of common variables (SNPs) with it. The aggregation matrix is formed using the matrix of interactions from this subset of problems. Information about the interactions of problem i is not included in the matrix.
- Limited information, EDA results ($Tn4$): Similarly to $Tn3$, information about only a very reduced subset of the SNP databases is used. However, the subset of the problems most related to i is the one from which Tree-EDA^r takes the most profit in terms of optimization results. Tree-EDA^r is initially run for problem i using in every run only the interactions between variables in i that are in problem j . Then the interactions of the subset of problems the information of which gave the best optimization results for problem i are aggregated. Information about the interactions of problem i is not included in the matrix. This way we assume that aggregating information from problems that have independently helped to reach good solutions should improve the optimization results.

In the section of experiments we present the results for these implementations.

VI. RELATED WORK

It is a common practice in EDAs to use available information about the problem to improve the efficiency of the learning and sampling steps of the algorithms. This can be achieved in a variety of ways: Using the known structural information to define a factorization of the probabilistic model [21], [22], constraining the set of interactions to be included in the probabilistic model [23], [24], and specifying soft constraints to bias the construction of the probabilistic model [25].

One of the distinguished features of our method is that we consider transfer of information between problems that are only partially related. It has been acknowledged that biasing

model learning using approaches such as the probability coincidence matrix [25] are only applicable when the structure of the underlying problem does not change much between different problem instances. The different approaches presented in this paper for constructing aggregation matrices prove that transfer learning can be applied between problems domains that share a minor, although to some extent complementary, relationship. On the other hand, although we have focused on information that is available a priori, we could extract similar information from the trees generated and construct aggregation matrices from this information. Finally, using the EDA performance as a source of information for matrix aggregation is a different way to bias model learning in EDAs.

Recently [26], [27], network measures extracted from the graphical structures of the probabilistic graphical models have been applied for transferring knowledge between problems and algorithms in EDAs. However, these results were achieved for problem instances having the same number of variables and results on transfer learning were constrained to the use of the information collected along the EDA evolution for classifying and predicting problem features.

VII. EXPERIMENTS

A. Experimental setup

The main objective of our experiments is to show that the use of structural transfer can improve the results for the multi-marker tagging SNP selection problem. We also show that the use of interactions is essential for the solution of this problem and that our proposed methods to bias the construction of the probabilistic models can produce gains in terms of solution quality and efficiency.

1) *Description of the SNP problem benchmark:* We use the SNP dataset previously applied in [28] that comprises genotypes of 1167 unrelated individuals from 58 populations. 927 individuals from the HGDP-CEPH cell line panel [29], 30 individuals who had previously been included in an investigation of microsatellite variation in Indian populations [30] and the other from HapMap [31] (60 from CEU, 60 from YRI and 90 from CHB+JPT). The number of SNPs was 1653.

2) *Preprocess:* Given the data set D consisting of a population of m haplotypes, first we remove those SNPs for which the frequency of the most probable allele is above 0.95. Then we compute the r_{ij}^2 for each pair of SNPs s_i and s_j , and afterward $r_{\{i,j\},k}^2$ is computed for $i \neq j \neq k$. Only pairs of SNPs that are in the sequence at a distance lower than $d = 40000$ are considered. The resulting set of all initial pairs and triples is reduced by eliminating those subsets of SNPs with an r^2 below the minimum threshold $r_{min}^2 = 0.8$.

Table I shows a description of the SNP sets used in our experiments. In that table, *Population* is the human population from which SNP samples were extracted; *n. ind.* is the number of human subjects from which samples were collected; *SNPs* is the number of SNPs that passed the minor allele frequency (MAF) threshold; n' is the number of SNPs that are not self-tagging which corresponds to the number of variables of the optimization problem; *pairs* is the number of SNP

<i>Index</i>	<i>Population</i>	<i>n.ind.</i>	<i>SNPs</i>	n'	<i>pairs</i>	<i>triples</i>
1	<i>Adygei</i>	15	1388	1048	3445	203640
2	<i>Balochi</i>	24	1399	1020	3486	220892
3	<i>BantuKenya</i>	11	1338	800	2183	89372
4	<i>BantuSouthAfrica</i>	8	1379	905	2553	127780
5	<i>Basque</i>	24	1362	1042	3956	252296
6	<i>Bedouin</i>	45	1433	1015	3147	182836
7	<i>Bengali</i>	15	1390	1007	3374	211439
8	<i>BiakaPygmy</i>	23	1322	700	2071	78136
9	<i>Brahui</i>	24	1368	975	2770	140452
10	<i>Burusho</i>	23	1388	1016	3272	197048
11	<i>CEU</i>	60	1386	1018	3227	191730
12	<i>Cambodian</i>	8	1335	1067	3476	219632
13	<i>Colombian</i>	7	1175	1008	4467	264906
14	<i>Dai</i>	10	1267	982	3762	212758
15	<i>Daur</i>	10	1248	956	3372	187016
16	<i>Druze</i>	41	1377	1019	3306	190120
17	<i>French</i>	28	1418	1068	3596	229216
18	<i>Han</i>	34	1273	946	3527	211194
19	<i>Han – NChina</i>	10	1230	954	3270	188553
20	<i>Hazara</i>	22	1329	968	2868	159534
21	<i>Hezhen</i>	9	1307	1016	4061	263278
22	<i>Italian</i>	12	1292	994	3322	194278
23	<i>JPT + CHB</i>	90	1295	938	3239	185792
24	<i>Japanese</i>	29	1296	983	3551	217660
25	<i>Kalash</i>	23	1346	1060	3344	197854
26	<i>Karitiana</i>	14	984	846	3413	165038
27	<i>Lahu</i>	8	1251	1012	3933	242766
28	<i>Makrani</i>	25	1400	1025	3267	194082
29	<i>Mandenka</i>	21	1338	792	2370	106988
30	<i>Maya</i>	21	1175	924	3275	178733
31	<i>MbutiPygmy</i>	12	1142	659	1937	76007
32	<i>Melanesian</i>	11	1100	875	3428	200288
33	<i>Miao</i>	10	1177	920	3163	177316
34	<i>Mongola</i>	10	1283	971	3290	187515
35	<i>Mozabite</i>	28	1475	1000	3102	177130
36	<i>Naxi</i>	8	1307	1040	3622	233508
37	<i>Orcadian</i>	13	1307	1022	3291	194862
38	<i>Oroqen</i>	9	1357	1089	3957	243990
39	<i>Palestinian</i>	44	1434	1006	3112	174598
40	<i>Papuan</i>	16	1092	860	3436	192654
41	<i>Pathan</i>	24	1422	1050	3442	213482
42	<i>Pima</i>	12	976	815	3117	155368
43	<i>Russian</i>	24	1341	1018	3422	199504
44	<i>San</i>	6	1087	735	2038	78260
45	<i>Sardinian</i>	27	1367	1045	3797	227492
46	<i>She</i>	10	1223	943	3353	192382
47	<i>Sindhi</i>	24	1416	1023	3285	198290
48	<i>Surui</i>	8	932	839	4247	188254
49	<i>Tamil</i>	15	1381	1021	3142	184799
50	<i>Tu</i>	10	1289	994	3643	213612
51	<i>Tujia</i>	10	1202	930	3148	168863
52	<i>Tuscan</i>	6	1320	1097	3945	235882
53	<i>Uygur</i>	10	1411	1077	3554	224617
54	<i>Xibo</i>	9	1331	1055	3821	243084
55	<i>YRI</i>	60	1405	780	2301	97496
56	<i>Yakut</i>	25	1302	988	3511	215246
57	<i>Yi</i>	10	1270	987	3620	211521
58	<i>Yoruba</i>	22	1346	790	2347	102510

TABLE I
DESCRIPTION OF THE SNPs SETS USED IN THE EXPERIMENTS.

pairs with a correlation above the 0.8 threshold and; similarly, triples is the number of SNP triples where two of the SNPs have a correlation above 0.8 with the third. The number of variables, pairs and triples all influence the complexity of the SNP tagging optimization problem.

<i>I</i>	<i>Best</i>			<i>mean</i>		
	<i>UMDA</i>	<i>Tree</i>	<i>TreeR</i>	<i>UMDA</i>	<i>Tree</i>	<i>TreeR</i>
1	362	285	285	367.63	291.30	<u>288.13</u>
2	365	302	300	371.17	304.50	<u>303.73</u>
3	344	284	283	349.73	285.97	<u>285.53</u>
4	372	305	306	378.47	308.67	<u>308.33</u>
5	354	283	279	360.80	286.53	<u>284.40</u>
6	376	299	298	381.63	302.37	<u>301.73</u>
7	341	276	273	346.93	281.27	<u>280.00</u>
8	325	265	264	329.23	265.70	<u>265.43</u>
9	364	296	294	374.53	299.37	<u>298.40</u>
10	351	288	288	359.30	<u>291.43</u>	291.50
11	360	293	294	370.00	299.43	<u>297.50</u>
12	341	270	270	346.93	276.00	<u>273.97</u>
13	286	235	233	294.20	238.90	<u>236.30</u>
14	325	263	261	332.37	266.90	<u>263.13</u>
15	325	264	264	332.90	267.63	<u>267.67</u>
16	361	291	291	369.90	297.37	<u>296.50</u>
17	360	284	284	374.17	290.70	<u>289.07</u>
18	326	257	258	331.87	262.37	<u>261.50</u>
19	330	266	263	336.60	269.43	<u>268.00</u>
20	361	292	290	366.77	296.37	<u>294.00</u>
21	327	256	256	332.33	263.27	<u>260.70</u>
22	336	274	273	340.97	278.23	<u>277.27</u>
23	322	259	255	327.50	261.23	<u>260.00</u>
24	354	278	275	360.00	282.43	<u>280.80</u>
25	365	287	287	371.63	294.70	<u>292.60</u>
26	246	199	200	253.07	203.87	<u>203.67</u>
27	319	251	251	322.50	254.70	<u>254.93</u>
28	361	293	292	365.60	296.30	<u>294.23</u>
29	323	270	269	329.83	273.03	<u>273.00</u>
30	303	239	236	308.00	243.47	<u>241.30</u>
31	269	229	228	273.70	230.13	<u>230.27</u>
32	280	222	224	284.53	228.20	<u>227.07</u>
33	315	247	247	321.17	252.07	<u>250.17</u>
34	329	264	260	332.77	266.93	<u>264.37</u>
35	369	294	294	377.00	297.67	<u>297.13</u>
36	343	274	271	349.67	278.80	<u>276.43</u>
37	349	280	281	355.00	285.03	<u>284.60</u>
38	347	281	281	355.33	<u>285.70</u>	287.67
39	372	300	298	381.90	305.30	<u>304.87</u>
40	285	226	228	292.13	<u>230.53</u>	230.53
41	355	284	285	362.63	288.80	289.07
42	237	189	190	240.93	<u>193.53</u>	193.67
43	348	279	280	355.40	282.30	283.47
44	259	230	231	267.40	<u>232.43</u>	232.47
45	362	284	282	369.10	287.50	287.90
46	316	252	252	323.37	255.93	<u>255.63</u>
47	360	290	292	367.17	296.80	<u>297.67</u>
48	227	184	183	232.87	188.13	<u>187.37</u>
49	362	284	282	368.27	287.80	<u>287.33</u>
50	329	264	264	338.10	<u>267.30</u>	267.73
51	331	258	261	337.17	<u>264.13</u>	264.47
52	359	283	283	365.80	288.13	<u>287.90</u>
53	373	295	295	379.57	299.13	299.57
54	351	277	278	355.97	281.80	<u>281.43</u>
55	329	265	266	335.17	268.30	<u>268.03</u>
56	340	265	261	344.63	268.23	<u>264.53</u>
57	619	556	551	624.93	559.07	<u>557.20</u>
58	325	269	270	329.53	274.07	<u>273.13</u>

TABLE II

RESULTS OF THE COMPARISON BETWEEN THE DIFFERENT EDAs.

<i>I</i>	<i>Best</i>				<i>mean</i>			
	<i>Tn1</i>	<i>Tn2</i>	<i>Tn3</i>	<i>Tn4</i>	<i>Tn1</i>	<i>Tn2</i>	<i>Tn3</i>	<i>Tn4</i>
1	281	281	280	283	285.60	<u>285.20</u>	287.00	287.33
2	296	298	299	297	<u>300.13</u>	300.97	302.93	302.00
3	281	282	283	284	<u>284.27</u>	284.60	285.80	285.13
4	304	304	306	306	306.47	<u>306.17</u>	308.60	309.03
5	276	278	282	282	281.53	<u>281.47</u>	286.93	286.00
6	295	297	295	294	299.23	<u>298.97</u>	300.50	299.00
7	271	269	271	271	275.60	<u>275.00</u>	277.97	275.80
8	263	263	265	264	265.07	<u>264.63</u>	266.43	265.73
9	291	293	291	292	<u>294.80</u>	295.73	295.47	296.17
10	284	285	283	284	<u>287.33</u>	288.20	287.80	287.80
11	291	291	289	288	296.50	295.67	295.33	<u>294.23</u>
12	265	266	270	269	<u>271.07</u>	271.70	274.67	274.20
13	230	229	229	232	233.57	<u>233.47</u>	235.83	236.80
14	257	255	260	258	<u>261.03</u>	261.27	263.37	262.83
15	260	261	266	261	264.73	<u>264.47</u>	268.17	264.97
16	290	286	289	288	293.63	<u>292.17</u>	292.40	293.60
17	281	279	282	284	<u>285.80</u>	286.10	287.27	288.63
18	255	255	255	256	<u>258.13</u>	258.43	258.63	259.17
19	261	262	260	262	<u>265.90</u>	266.30	266.03	267.00
20	287	288	289	291	292.40	<u>292.03</u>	293.33	295.03
21	253	254	254	253	<u>257.27</u>	257.27	258.77	257.67
22	270	270	272	271	274.53	274.67	275.80	<u>274.47</u>
23	253	254	255	254	257.10	<u>256.43</u>	257.93	257.00
24	271	272	274	272	277.37	<u>277.20</u>	278.97	277.33
25	284	284	290	289	288.77	<u>288.13</u>	293.70	293.70
26	197	195	199	201	200.70	<u>200.10</u>	201.83	203.83
27	247	245	246	248	250.03	<u>249.70</u>	252.90	252.27
28	287	289	293	290	<u>291.93</u>	292.20	295.67	293.97
29	269	269	270	270	<u>271.17</u>	271.50	271.60	272.10
30	233	235	236	240	<u>238.77</u>	240.17	240.10	242.77
31	227	229	229	231	<u>229.27</u>	230.63	230.43	233.10
32	220	220	223	223	224.80	<u>223.90</u>	226.90	226.83
33	243	244	242	244	<u>247.33</u>	248.07	247.87	248.43
34	257	257	261	258	262.50	<u>261.73</u>	264.20	262.87
35	291	292	292	294	<u>295.30</u>	296.10	295.63	297.77
36	268	270	269	270	273.63	274.13	<u>272.73</u>	275.93
37	277	278	278	280	281.77	<u>281.53</u>	281.70	284.40
38	277	278	281	282	<u>281.77</u>	282.77	285.63	286.50
39	298	297	299	297	<u>301.77</u>	301.50	303.67	302.67
40	222	224	225	223	<u>226.40</u>	227.60	229.17	226.53
41	278	281	279	282	<u>283.60</u>	283.70	285.63	285.57
42	186	185	188	188	189.40	<u>189.33</u>	190.87	190.60
43	275	273	277	275	279.40	<u>278.87</u>	281.00	279.50
44	229	230	230	229	231.17	231.43	232.90	<u>230.77</u>
45	277	277	279	281	282.13	<u>282.10</u>	282.73	284.43
46	245	248	250	250	<u>250.20</u>	251.13	253.33	254.10
47	289	288	290	289	293.00	292.87	293.67	292.63
48	178	179	179	182	<u>183.00</u>	184.13	185.27	185.53
49	280	279	282	281	283.57	<u>283.37</u>	285.97	286.50
50	257	261	262	260	<u>262.87</u>	264.13	266.43	264.77
51	255	255	259	257	<u>259.23</u>	259.60	261.37	260.20
52	277	277	281	282	281.97	<u>281.40</u>	285.93	286.33
53	290	291	294	294	<u>293.77</u>	294.93	297.43	297.37
54	271	273	274	274	<u>276.43</u>	277.13	280.77	278.07
55	264	265	265	267	<u>267.10</u>	<u>266.70</u>	267.67	270.50
56	256	257	258	260	<u>261.30</u>	261.93	263.77	264.93
57	550	550	555	554	554.30	<u>553.90</u>	558.57	557.40
58	269	269	270	271	271.43	<u>271.40</u>	272.57	273.07

TABLE III

RESULTS OF THE DIFFERENT VARIANTS OF STRUCTURAL TRANSFER.

3) *Parameters of the algorithms:* The population size of Tree-EDA and Tree-EDA^r was set to 5,000 and the number of generations was set to 100. Truncation selection with parameter $T = 15(\%)$ was employed. In this selection scheme, the best $T * N$ individuals of the population are selected to construct the probabilistic model. We apply a replacement

strategy called best elitism in which the selected population at generation t is incorporated into the population of generation $t + 1$, keeping the best individuals found so far and avoiding to reevaluate their fitness function. The algorithm will stop when the maximum number of generations is reached or the selected population has become too homogeneous (no more

than 10 different individuals).

We conduct for each SNP problem a set of 30 experiments and extract statistical information from the analysis of these experiments. The performance of Tree-EDA was evaluated considering the fitness of the best and average solutions found in all the experiments.

B. Analysis of the problem structure

As a first step we investigate the similarity between the different optimization problems in terms of the variables involved. We compute the distances between each pair of populations using the number of common variables as a measure of relatedness. Figure 1 shows the relatedness between problem domains computed according to the number of variables. Similarly, Figure 2 shows the relatedness between problem domains computed according to their structural similarity. In this case, we compute the distance between the matrix of interactions for each pair of populations. The distance is the number of interactions where the matrices do not coincide. Notice that both measures of relatedness are associated. If two domains have few variables in common, then the number of shared interactions will be low.

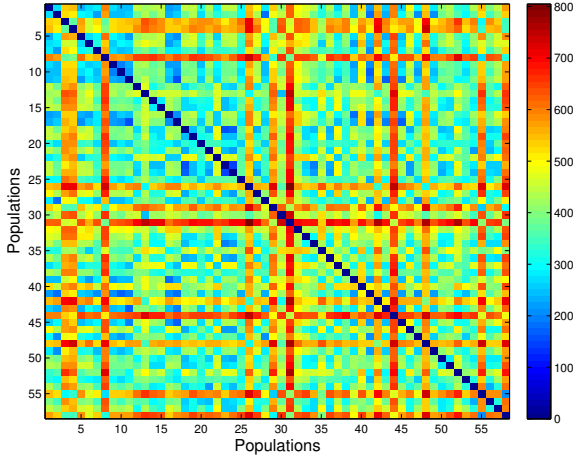


Fig. 1. Relatedness between problem domains according to the number of variables.

C. EDA numerical results

Our first experiment evaluates the importance of using variables interactions for the search of optimal solutions and the improvement produced by using information about the problem structure in the search. The use of structural transfer should be worth only if modeling the interactions between the problem variables is essential for solving the optimization problem. We compare Tree-EDA and Tree-EDA^r to the univariate marginal distribution algorithm (UMDA) [3], an EDA that models all the variables independently.

Table II shows the results of the algorithms. In the table, *I* refers to index of the optimization problem, *Best* is the best solution achieved by the EDA out of 30 experiments, *mean* is the mean value of the best solutions. The best mean

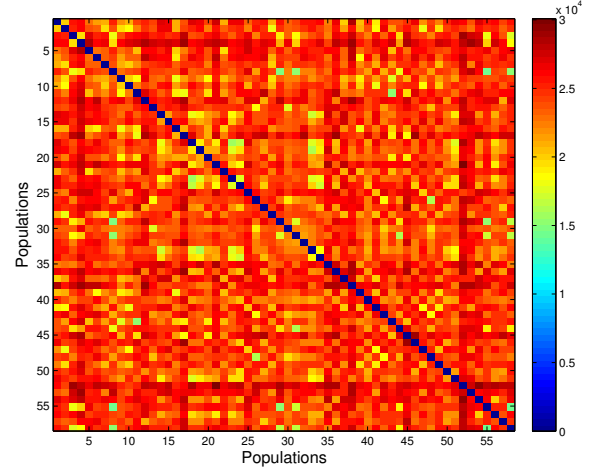


Fig. 2. Relatedness between problem domains according to the number of common interactions.

result for each problem is underlined. UMDA achieves the worst results for all the problems. On the other hand, the use of problem information as done by Tree-EDA^r improves the average results of Tree-EDA in 44 of the 58 problems.

D. Numerical results on transferability

In the next experiments we evaluate the performance of Tree-EDA^r using the different strategies for structural transfer. The number of problems used for computing aggregation matrices for variants *Tn3* and *Tn4* was 5 out of 58. Results are shown in Table III. In the table, the best results achieved among the four variants is underlined. Using *Tn1* the original results of Tree-EDA and Tree-EDA^r (Table II) are improved for all the 58 problems. *Tn2* improves the results of the two EDAs for all problems except one. *Tn3* improves the results of Tree-EDA and Tree-EDA^r for the 91% and 79% of the problems, respectively. Finally, *Tn4* produces improvements of Tree-EDA and Tree-EDA^r for the 90% and 76% of the problems.

In general, the results of the transfer strategies are quite remarkable. Not surprisingly, aggregating the structural information from all SNP datasets available (*Tn1*) and from all the problems except the one being optimized (*Tn2*) produce more noticeable improvements that relying only on the subset of the 5 most similar problems (*Tn3* and *Tn4*). It is encouraging that a simple similarity measure as the number of common variables between problems (*Tn3*) can respectively produce improvements in Tree-EDA and Tree-EDA^r of 91% and 79%. This example clearly shows that an approximate measure of relatedness, and information from a very limited set of related problems, can be very effective to improve the solutions achieved by EDAs in certain classes of optimization problems.

VIII. CONCLUSION

In this paper we have proposed a general approach for transfer learning in EDAs. We have implemented different variants of EDAs that transfer information between related

problems by means of aggregation matrices. Our proposals have been applied to multi-marker tagging SNP selection, a problem where EDAs have achieved state of the art results [4]. We have shown that the use of partial information extracted from a set of related problems can not only reduce the computational burden associated with probabilistic model learning in EDAs but also improve the quality of the final results. We suggest the use of EDAs in a more general context where different optimization problems are characterized in terms of the structural relationships among their components. Common structural relationships are identified and conveniently transferred to improve the optimization results for related problems.

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